

A smooth double-cone reduction for Question 4.12 of Guerra–Raita–Schrecker

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Status. This is a partial result packet. It proves the smooth double-cone version of Question 4.12 from Guerra–Raita–Schrecker, but it does not settle the rough BV^{Div} formulation discussed in the same section.

Source. The source paper is Andre Guerra, Bogdan Raita, Matthew Schrecker, *Compensation phenomena for concentration effects via nonlinear elliptic estimates*, arXiv:2112.10657. Question 4.12 appears on page 28 of the source PDF:

no estimate for fields constrained to take values in $\text{Sym}_2 \cap \{\det \leq 0\}$. Note that $\mathcal{K}_2^- \cap \text{Sym}_2$ is a two-dimensional half-space, while $\tilde{A}$ takes values in the full linear space containing this half-space. Thus entrywise constraints are in some sense necessary for an estimate.

4.4. Density of smooth functions and separation. The previous two remarks provide examples of matrix fields A which take values in $\text{Sym}_2 \cap \{\det \leq 0\}$ and which falsify the inequalities of this section. Notice also that, in both cases, the matrix field A is *constant* on the boundary of the unit ball. A question which the previous remarks leave open is the following:

Question 4.12. *Can the inequalities of this section hold if we constrain the fields to take values in the double cone*

$$\text{Sym}_2 \cap \{\det \geq 0\} = \text{Sym}_2 \cap (\mathcal{K}_1^+ \cup \mathcal{K}_3^+) = \text{Sym}_2^+ \cup \text{Sym}_2^-?$$

In fact, for fields satisfying the exact constraint $\text{Curl } A = 0$, we have the following rigidity result, which is a particular case of a theorem of Šverák [69], see also the remarkable generalisations due to Faraco–Székelyhidi [27] as well as [71, 72, 73] for related results:

Proposition 4.13. *Let $A \in L^2(\mathbb{B}^2, \text{Sym}_2)$ be such that $\text{Curl } A = 0$ and $A|_{\mathbb{S}^1}$ is constant. Then*

$$\text{either } A \in \text{Sym}_2^+ \text{ a.e. in } \Omega \quad \text{or } A \in \text{Sym}_2^- \text{ a.e. in } \Omega.$$

Transcription of the target question. The question asks whether the inequalities of the two-dimensional section can hold when the fields are constrained to the double cone

$$\text{Sym}_2 \cap \{\det \geq 0\} = \text{Sym}_2^+ \cup \text{Sym}_2^-.$$

Notation. If $A = (a_{ij})$ is a 2×2 matrix field, let A_i be its i -th row and let

$$d_i(A) = \text{div } A_i = \partial_1 a_{i1} + \partial_2 a_{i2}.$$

All matrix norms below are equivalent; in the proof we use the Frobenius norm.

Proof Intuition

For a smooth field in the double cone, moving from the positive branch Sym_2^+ to the negative branch Sym_2^- can only happen through the zero matrix, because trace zero inside the double cone forces both eigenvalues to be zero. Therefore the branch-flip

$$B = \text{sgn}(\text{tr } A)A$$

creates a positive semidefinite field without adding an interface divergence term. Since $\det(-A) = \det A$ in dimension two, the determinant is unchanged. The source paper's positive-cone determinant inequality then applies to B .

Main Result

Theorem 1 (Smooth double-cone determinant estimate). *Let $A \in C_c^\infty(\mathbb{R}^2, \text{Sym}_2^+ \cup \text{Sym}_2^-)$. Then*

$$\int_{\mathbb{R}^2} \det A \, dx \leq \|d_1(A)\|_{L^1(\mathbb{R}^2)} \|d_2(A)\|_{L^1(\mathbb{R}^2)}.$$

Proof. Put $\tau = \text{tr } A$, and define

$$B(x) = \begin{cases} A(x), & \tau(x) \geq 0, \\ -A(x), & \tau(x) < 0. \end{cases}$$

Equivalently, $B = \text{sgn}(\tau)A$, with any value assigned to $\text{sgn}(0)$, since $\tau = 0$ in $\text{Sym}_2^+ \cup \text{Sym}_2^-$ implies $A = 0$. Thus $B(x) \in \text{Sym}_2^+$ for all x , and $\det B = \det A$.

We first record that B is Lipschitz. Indeed, the map $\Phi(M) = M$ on Sym_2^+ and $\Phi(M) = -M$ on Sym_2^- is Lipschitz on the double cone. If two matrices lie on the same branch this is immediate. If $M, P \in \text{Sym}_2^+$, then

$$\|M - P\|_F^2 = \|M\|_F^2 + \|P\|_F^2 - 2 \text{tr}(MP) \leq \|M\|_F^2 + \|P\|_F^2 + 2 \text{tr}(MP) = \|M + P\|_F^2,$$

because $\text{tr}(MP) \geq 0$. This gives the cross-branch Lipschitz bound.

It remains to identify the divergence. Let $\eta_\varepsilon(t) = t(t^2 + \varepsilon^2)^{-1/2}$ and $B^\varepsilon = \eta_\varepsilon(\tau)A$. Then $B^\varepsilon \rightarrow B$ in L^1 , and

$$\partial_j B_{ij}^\varepsilon = \eta_\varepsilon(\tau) \partial_j A_{ij} + \eta'_\varepsilon(\tau) A_{ij} \partial_j \tau.$$

For matrices in the double cone, $|A_{ij}| \leq |\tau|$. Hence

$$|\eta'_\varepsilon(\tau) A_{ij} \partial_j \tau| \leq C |\partial_j \tau|,$$

and the left-hand side tends pointwise to 0 wherever $\tau \neq 0$, while it is zero where $\tau = 0$ after the factor $|A_{ij}| \leq |\tau|$. Dominated convergence gives

$$\partial_j B_{ij} = \text{sgn}(\tau) \partial_j A_{ij}$$

in distributions, and therefore

$$d_i(B) = \text{sgn}(\tau) d_i(A), \quad \|d_i(B)\|_1 \leq \|d_i(A)\|_1.$$

The field B is compactly supported, Lipschitz, and positive semidefinite. Since $\text{Sym}_2^+ \subset \{a_{11}, a_{22} \geq 0\}$, Theorem 4.1 of the source paper, with the rough-field extension in Remark 4.3, yields

$$\int_{\mathbb{R}^2} \det B \, dx \leq \|d_1(B)\|_1 \|d_2(B)\|_1.$$

Using $\det B = \det A$ and the divergence bound proves the claim. $\square$

Corollary 1 (Smooth L^2 consequence). *Let $\mathcal{C} \subset \text{Sym}_2^+ \cup \text{Sym}_2^-$ be a closed cone such that*

$$|M|^2 \leq K \det M \quad (M \in \mathcal{C})$$

for some $K < \infty$. Then every $A \in C_c^\infty(\mathbb{R}^2, \mathcal{C})$ satisfies

$$\|A\|_{L^2(\mathbb{R}^2)} \leq C_K \|\text{Div } A\|_{L^1(\mathbb{R}^2)}.$$

Proof. The cone assumption and the theorem give

$$\|A\|_2^2 \leq K \int_{\mathbb{R}^2} \det A \leq K \|d_1(A)\|_1 \|d_2(A)\|_1 \leq C_K \|\text{Div } A\|_1^2.$$

□

Limitations and Review Notes

The result should be read as a smooth answer. It does not resolve the rough BV^{Div} problem discussed near Question 4.12. For a rough field with a jump directly from a positive semidefinite value to a negative semidefinite value, the transformation $A \mapsto \text{sgn}(\text{tr } A)A$ creates an additional jump term in the divergence. This is precisely the mechanism behind the density obstruction in the source paper.

The main point for human review is the distributional identity

$$\text{Div}(\text{sgn}(\text{tr } A)A) = \text{sgn}(\text{tr } A) \text{Div } A$$

for smooth double-cone-valued A . The approximation by $\eta_\varepsilon(\text{tr } A)$ is included above to make the vanishing of the sign-derivative term explicit.

Novelty Check

Bounded checks on 2026-07-03 searched the run indexes, the local parsed arXiv corpus, and public web search for exact phrases including *Question 4.12*, *double cone*, $\text{Sym}_2^+ \cup \text{Sym}_2^-$, *constrained Ornstein*, and the source arXiv id. No separate later answer was found; exact hits led back to the source paper.

References

- [1] A. Guerra, B. Raita, and M. Schrecker, *Compensation phenomena for concentration effects via nonlinear elliptic estimates*, Ars Inveniendi Analytica (2024), Paper No. 1; arXiv:2112.10657.